Invention Disclosure Details

Name and Address of APPLICANT:

MANIPAL UNIVERSITY JAIPUR	Jaipur-Ajmer Express Highway, Dehmi Kalan, Near GVK Toll Plaza, Jaipur, Rajasthan 303007

Name and Address of Inventor:

Name	Department/Address
Mahir Desai	Department of Computer Science and Engineering, School of Computer Science and Engineering, Manipal University Jaipur
Kappa Geervan	Department of Computer Science and Engineering, School of Computer Science and Engineering, Manipal University Jaipur
Juhi Singh	Department of Computer Science and Engineering, School of Computer Science and Engineering, Manipal University Jaipur

DETAILS OF THE PUBLICATION/COMMUNICATION/EXHIBITION IN WHICH THE INVENTION WAS DISCLOSED PRIOR TO PATENT FILING IN PUBLIC DOMAIN. (If YES kindly provide details)

Response: NO

If yes mention the details and provide the link.

TITLE: System and Method for Adaptive Learning from Presentation Slides Using Multi-Modal Content Generation and Concept Graph Modeling

ABSTRACT

This invention discloses a system and method for transforming presentation slides into an adaptive, multi-modal learning environment using artificial intelligence. The system processes presentation content to extract structured information and generates a unified Slide Intelligence representation comprising summaries, key concepts, and assessment elements.

A Concept Dependency Graph is constructed to model relationships between extracted concepts, enabling prerequisite-aware learning and structured knowledge progression. An adaptive scheduling mechanism dynamically selects appropriate learning modes based on user interaction data and performance metrics.

The system further incorporates a feedback-driven validation loop that verifies structural correctness and pedagogical consistency of generated content, with iterative refinement applied when constraints are not satisfied. Additionally, a deterministic caching mechanism minimizes redundant computation and ensures consistency across repeated requests.

The invention provides a scalable and intelligent framework for personalized learning from presentation-based educational materials.

Problem Solved by the Invention

Presentation slides are widely used for educational and professional purposes; however, they inherently lack interactivity, assessment mechanisms, and adaptive learning capabilities. Existing solutions typically provide isolated functionalities such as summarization or conversational tutoring, but fail to integrate structured knowledge extraction, concept relationship modeling, and adaptive learning progression within a unified system.

Furthermore, current approaches do not effectively capture prerequisite dependencies between concepts or dynamically adapt learning content based on individual user performance. The repeated generation of AI-based outputs also introduces variability and inconsistency in learning materials.

The present invention addresses these limitations by integrating slide content extraction, concept dependency graph modeling, adaptive mode selection, feedback-driven validation, and deterministic caching into a cohesive framework, thereby enabling consistent, structured, and personalized learning experiences.

Brief Literature Review & How This Invention Differs

The following prior art documents represent relevant developments in adaptive learning systems, concept modeling, and AI-based educational platforms:

1. US20210027647A1 – Adaptive Machine Learning System (2021)

Link: https://patents.google.com/patent/US20210027647A1

This patent discloses an adaptive learning system that monitors user interactions and dynamically optimizes learning paths using machine learning techniques.

Difference:

The system does not disclose processing of presentation slides or extraction of structured content from slide formats. It fails to teach generation of multiple pedagogical learning modes from a single content source and does not construct concept dependency graphs derived from instructional material. Furthermore, it does not disclose deterministic caching or feedback-driven validation mechanisms.

2. US20250378766A1 – Systems, Methods, and Apparatuses for Implementing an Adaptive and Scalable AI-Driven Personalized Learning Platform (Arizona State University, 2025)

Link: https://patents.google.com/patent/US20250378766A1

This patent describes a personalized learning platform that maintains learner profiles using graph-based representations and generates contextual responses using AI models.

Difference:

The system is limited to learner modeling and content delivery and does not disclose slide-based content extraction or transformation into structured learning representations. It fails to teach multi-modal learning generation from a single input source and does not construct concept dependency graphs derived from slide content. Additionally, it does not disclose deterministic caching or validation loops.

3. US20220415199A1 – Smart-Learning and Knowledge Concept Graphs (2022)

Link: https://patents.google.com/patent/US20220415199A1

This patent discloses a system that generates personalized concept graphs based on user queries and multiple data sources.

Difference:

The system is limited to concept graph generation and does not disclose slide-based knowledge extraction or transformation into structured learning representations. It fails to teach multi-modal learning generation and does not integrate adaptive scheduling, deterministic caching, or feedback-driven validation.

4. US12430517B2 – Method and System for Document Structure-Based Unsupervised Long-Form Technical Question Generation (Tata Consultancy Services, 2025)

Link: https://patents.google.com/patent/US12430517B2

This patent extracts document structure and generates technical questions using predefined templates.

Difference:

The system is limited to question generation and does not disclose generation of multiple learning modes such as memorization, visualization, or guided tutoring. It fails to teach concept dependency graph construction or adaptive learning based on user performance. Additionally, it does not disclose caching or validation mechanisms.

5. US20220237548A1 – Gamified Real-Time Artificial Intelligence Based Individualized Adaptive Learning (2022)

Link: https://patents.google.com/patent/US20220237548A1

This patent discloses a gamified adaptive learning system that personalizes learning paths using behavioral data and engagement strategies.

Difference:

The system focuses on engagement optimization and does not disclose slide-based content extraction or structured representation. It fails to teach multi-modal learning generation, concept dependency modeling, deterministic caching, or feedback-driven validation of generated content.

Summary of Differentiation

Existing systems independently address adaptive learning, concept graph modeling, or automated content generation. However, none disclose an integrated framework that combines slide-based structured content extraction, concept dependency graph construction, multi-modal learning generation, adaptive learning based on user performance, deterministic caching, and feedback-driven validation.

The present invention uniquely integrates these components into a unified system, enabling structured, adaptive, and consistent learning from presentation-based materials.

Advantages of the Invention

The present invention provides the following technical advantages over existing systems:

1. Automated Learning Content Generation

The system converts static presentation slides into structured learning representations without requiring manual creation of summaries, quizzes, or study material.

2. Graph-Based Learning Navigation

A Concept Dependency Graph enables prerequisite-aware learning, ensuring that users understand foundational concepts before progressing to advanced topics.

3. Adaptive Learning Mechanism

The system dynamically adjusts learning modes based on user performance, enabling personalized learning paths without manual configuration.

4. Efficient Caching for Reduced Computation

A deterministic Slide Intelligence Cache prevents repeated AI processing of identical content, improving response time and reducing computational cost.

5. Consistent Multi-Mode Learning

All learning modes operate on a shared structured representation, ensuring consistency across memorization, quizzes, visualization, and tutoring interactions.

6. Integrated Multi-Modal Learning from Single Source

Multiple learning formats are generated from a single slide input, eliminating the need for separate tools or pipelines.

7. Concept Gap Detection and Remediation

The system identifies gaps in understanding and redirects users to prerequisite content, improving learning efficiency.

Salient Features of the Invention

The present invention introduces a set of integrated technical features that collectively enable the transformation of static presentation slides into an adaptive, multi-modal learning system. The salient features distinguishing the invention from existing systems are as follows:

A. Unified Slide Intelligence Representation
The system generates a structured Slide Intelligence Object for each slide, comprising semantic summaries, key concepts, and assessment elements. This unified representation serves as a consistent knowledge layer for all learning modes, ensuring coherence across different forms of content delivery.

B. Deterministic Slide Intelligence Caching Mechanism
A content-based hashing technique is employed to store and retrieve Slide Intelligence Objects. This ensures that identical inputs produce consistent outputs, reduces redundant AI computations, and improves system efficiency across repeated interactions.

C. Concept Dependency Graph Derived from Slide Content
The system constructs a directed Concept Dependency Graph in which nodes represent extracted concepts and edges represent prerequisite relationships. This enables structured navigation of learning material and supports prerequisite-aware progression.

D. Concept Gap Detection and Remediation
The system evaluates user performance against prerequisite concepts in the dependency graph to identify knowledge gaps. When gaps are detected, the system adaptively redirects the learner to relevant foundational content before progressing further.

E. Adaptive Learning Mode Scheduler

An adaptive scheduling mechanism analyzes user performance metrics and dynamically selects appropriate learning modes. The system transitions between memorization, assessment, visualization, and guided explanation based on mastery levels and interaction history.

F. Cross-Mode Consistency via Shared Intelligence Layer
All learning modes operate on the same Slide Intelligence Object, ensuring that generated explanations, questions, and visual representations remain consistent across different modes of interaction.

G. Multi-Modal Learning Generation from a Single Content Source
The system generates multiple pedagogical formats—including summaries, quizzes, visual representations, and guided explanations—from a single slide input, eliminating the need for manual content creation across different formats.

H. Feedback-Driven Validation and Adaptation Loop
Generated content is validated against predefined structural and pedagogical constraints. In case of inconsistencies, the system iteratively refines the output. Additionally, user interaction data is continuously incorporated to adapt future content generation and learning pathways.

I. Modular Orchestration with Iterative Refinement
The system employs coordinated processing modules—including classification, difficulty estimation, generation, and validation—that interact in a structured pipeline with iterative feedback, enabling adaptive and context-aware learning content generation.

METHODOLOGY ADOPTED and RESULTS OBTAINED

Methodology Adopted

The system follows a structured multi-stage processing pipeline integrated with a feedback-driven validation mechanism for adaptive and consistent content generation.

1. Slide Ingestion and Preprocessing

Presentation files (PPT/PDF) are processed to extract textual content while filtering non-informative elements. The extracted data is organized into slide-level units, preserving structural components such as titles, bullet points, and key textual elements.

2. Slide Intelligence Construction

Each slide is transformed into a structured representation referred to as a Slide Intelligence Object, comprising:

- Semantic summary
- Key concepts
- Assessment elements
- Metadata including content type and difficulty

This representation serves as the foundational knowledge layer for all subsequent processing.

3. Concept Dependency Graph Generation

A directed Concept Dependency Graph is constructed wherein nodes represent extracted concepts and edges represent prerequisite relationships. This enables structured learning progression and identification of conceptual dependencies.

4. Adaptive Learning Scheduling

User performance is evaluated using metrics such as response accuracy and interaction patterns. Based on these metrics, the system dynamically assigns appropriate learning modes, including memorization, assessment, visualization, or guided explanation.

5. Multi-Mode Learning Execution

The system generates multiple learning formats—including summaries, quizzes, visual representations, and guided interactions—from the Slide Intelligence Object. All modes operate on a shared representation to ensure consistency.

6. Deterministic Caching Mechanism

A content-based hashing mechanism is employed to store and retrieve generated outputs. This ensures consistent responses for identical inputs, reduces redundant processing, and improves system efficiency.

7. Feedback and Adaptive Update Mechanism

User interactions are continuously monitored to update learner profiles. This enables dynamic adjustment of difficulty levels, detection of knowledge gaps, and personalized learning progression.

8. Feedback-Driven Generative Validation Loop

Generated outputs are evaluated against predefined structural and semantic constraints. If validation fails, corrective feedback is provided to the generation module, and the content is iteratively refined until valid output is produced or a retry threshold is reached, ensuring reliability and consistency.

Results Obtained

A functional prototype of the proposed system has been implemented and evaluated using multiple presentation datasets.

The system successfully extracts structured content from PPT and PDF formats while preserving hierarchical elements such as titles and bullet points. The Slide Intelligence module generates consistent summaries, concepts, and assessment items, supported by a deterministic caching mechanism that reduces redundant computation and improves response time.

The Concept Dependency Graph effectively models relationships between concepts and supports identification of prerequisite dependencies. The system is able to detect conceptual gaps and guide users toward foundational topics when insufficient mastery is observed.

The adaptive learning mechanism dynamically selects appropriate learning modes based on user performance, enabling personalized learning progression. Lower mastery levels trigger reinforcement-oriented modes, while higher proficiency enables advanced interaction modes.

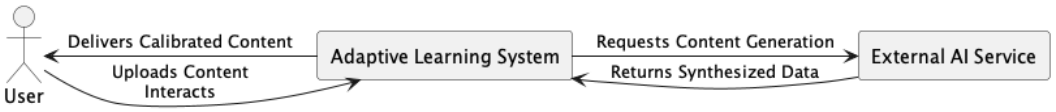
The use of a shared Slide Intelligence representation ensures consistency across multiple learning modes, eliminating variability in generated outputs.

Overall, the system demonstrates effective integration of slide content processing, concept graph modeling, adaptive learning, validation mechanisms, and caching, resulting in improved learning efficiency and reduced computational overhead.

FIGURES / Diagrams related to the invention with their title.

1. System Context Diagram

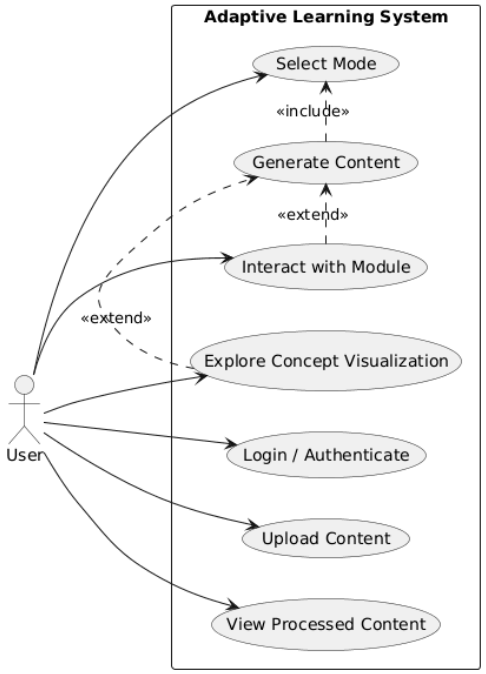
FIG. 1: System Context Diagram



Description: Depicts the interaction between the user and the adaptive learning system, including content upload, mode selection, and AI-driven content generation through external services.

2. Use Case Diagram

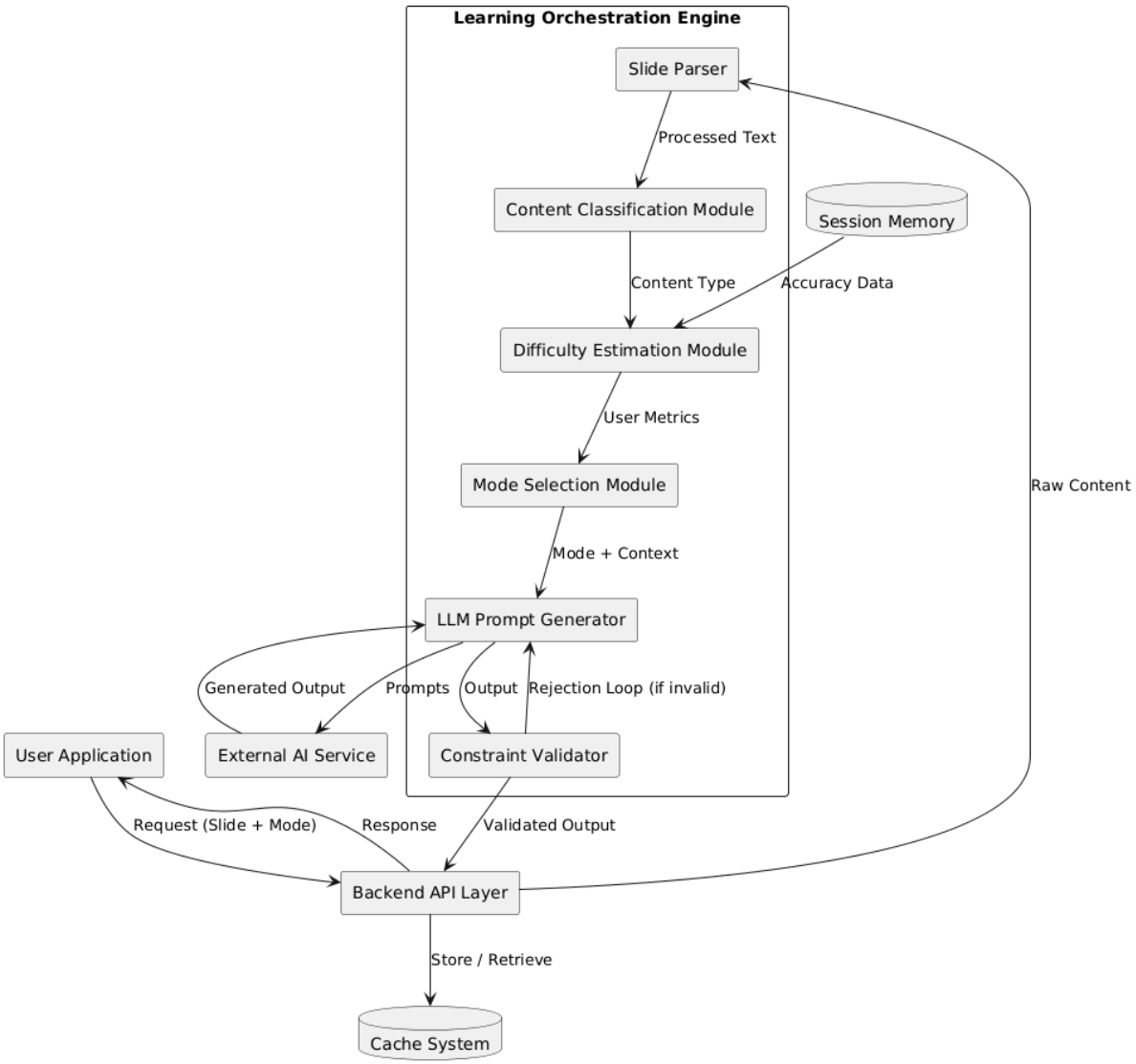
FIG. 2: Use Case Diagram



Description: Illustrates the core functionalities of the system including content upload, authentication, mode selection, concept visualization, and interaction with adaptive learning modules.

3. High-Level Architecture Diagram

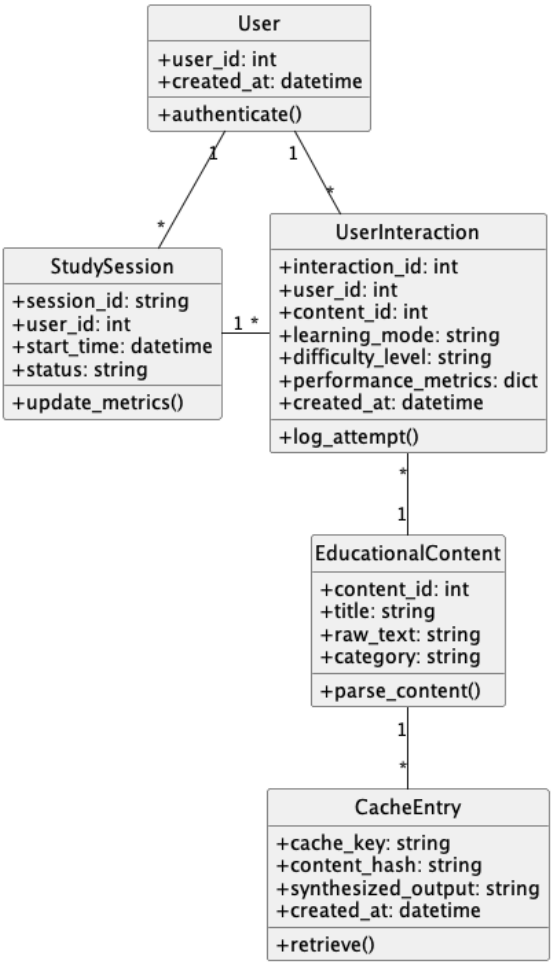
FIG. 3: High-Level Architecture of Adaptive Learning System



Description: Represents the modular architecture of the system including slide parsing, content classification, difficulty estimation, adaptive mode selection, LLM-based generation, validation, caching, and interaction with external AI services.

4. Class Diagram

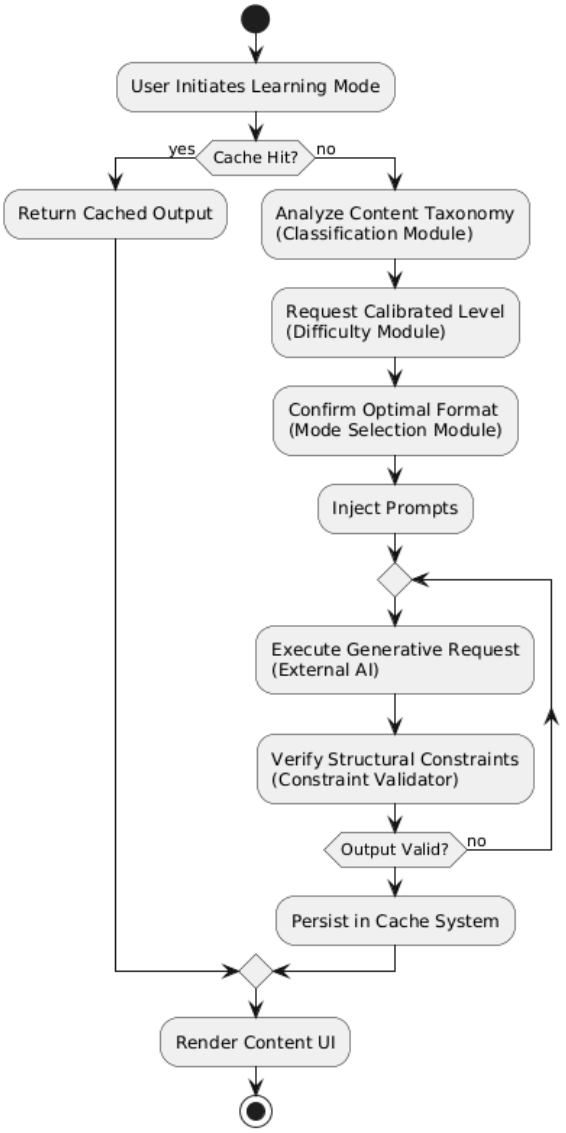
FIG. 4: Class Diagram



Description: Represents the system data model including entities such as user, study session, content, user interaction, and cache entries, along with their relationships.

5. Activity Diagram

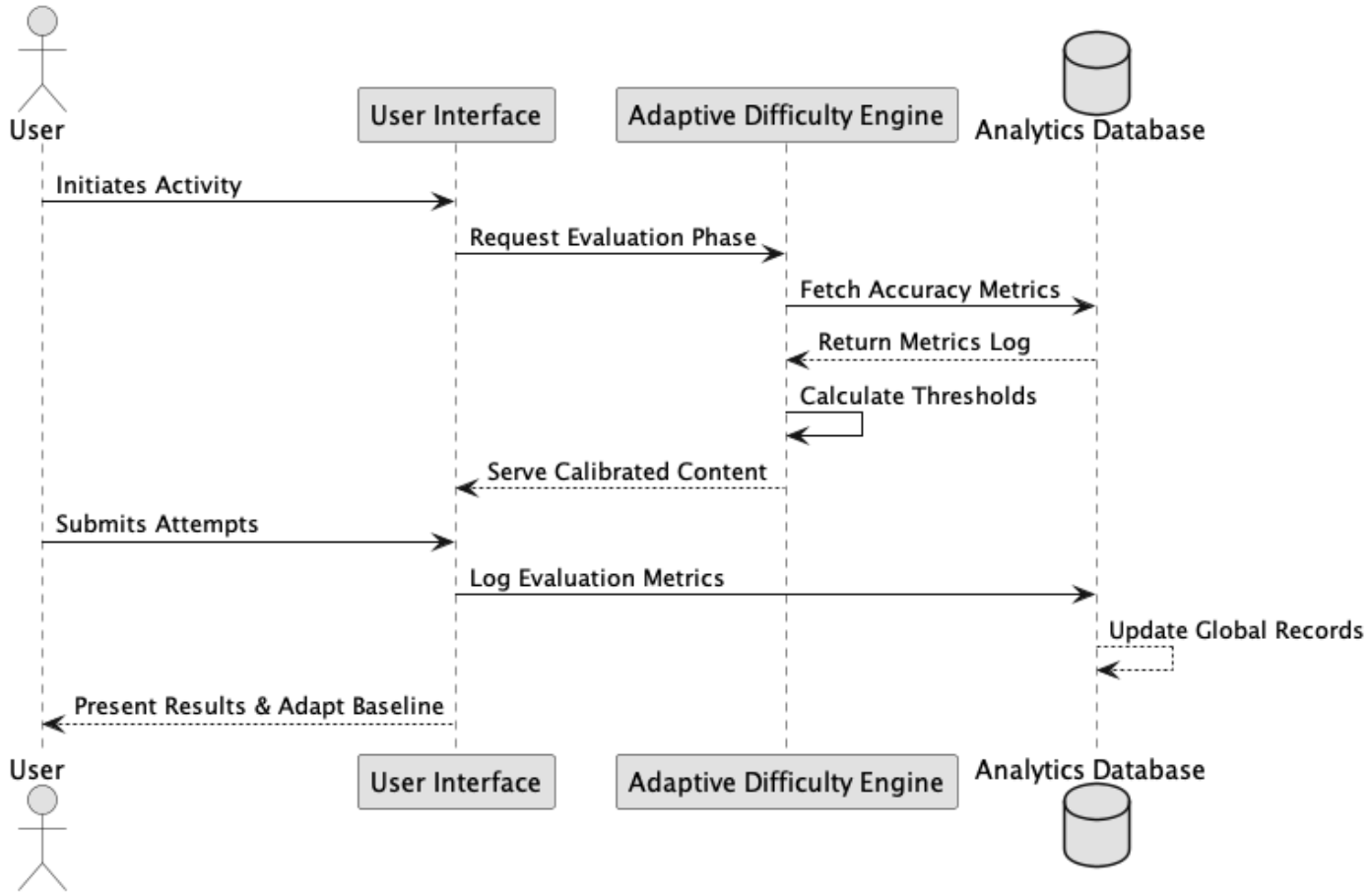
FIG. 5: Content Processing Flow (Activity Diagram)



Description: Describes the content processing workflow including cache lookup, classification, adaptive mode selection, AI generation, validation, and response delivery.

6. Sequence Diagram (Adaptive Learning)

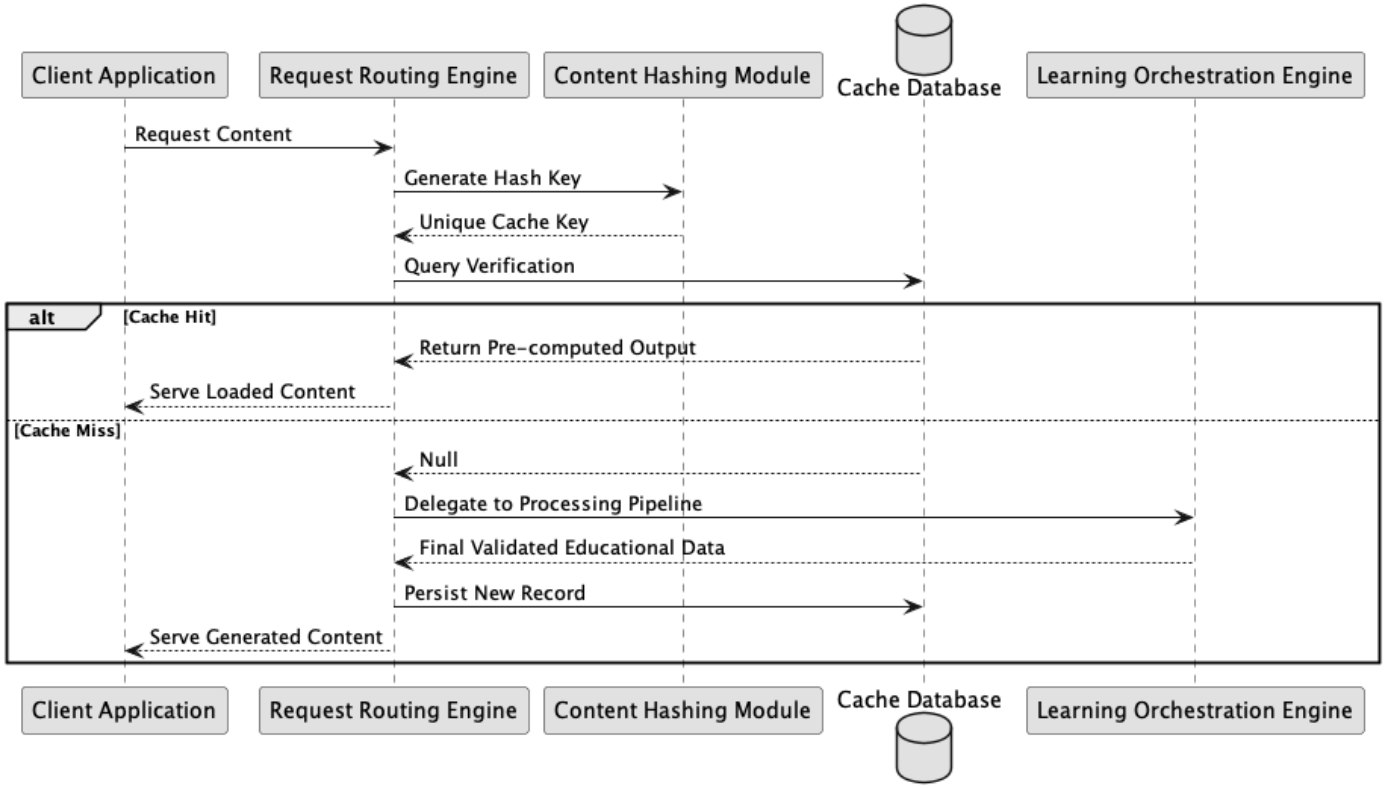
FIG. 6: Sequence Diagram – Adaptive Learning



Description: Illustrates the interaction between the user interface, adaptive difficulty engine, and analytics system for evaluating performance and dynamically adjusting learning difficulty.

7. Sequence Diagram (Cache System)

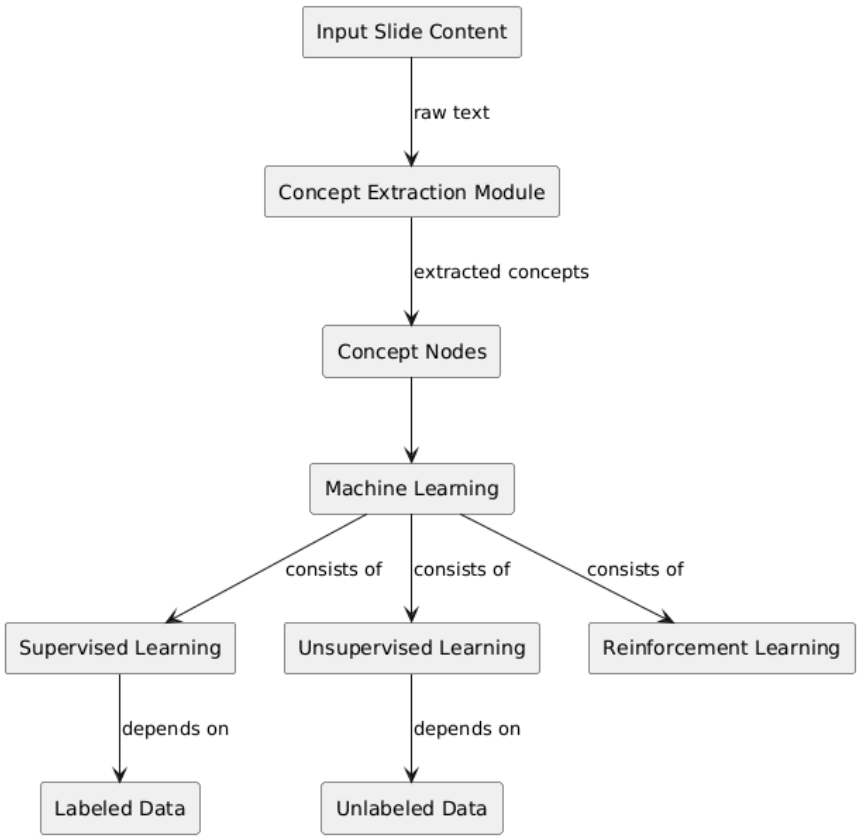
FIG. 7: Sequence Diagram – Cache Architecture



Description: Depicts the cache-based content retrieval mechanism including hashing, cache lookup, conditional AI execution, and storage of generated results.

8. Concept Dependency Graph

FIG. 8: Semantic Concept Dependency Graph



Description: Represents the extraction of concepts from slide content and the construction of dependency relationships to support structured and prerequisite-aware learning.